

The decision after acceptance rejection

Every stack in scope has failed required acceptance testing.

01 Acceptance result

The HBM3E stack is rejected. The diagnostic path does not change its disposition.

02 Diagnostic decision

The on-duty quality engineer chooses which of five diagnostic procedures to run next.

03 Process value

The findings identify fault mechanisms for process engineering. The stack will be scrapped either way.

Helion supports the investigation. The engineer owns the decision.

Source: Client brief, Business Context and Objective.

The investigation before a procedure

01 Load available records

Use manufacturing, package-inspection and acceptance observations available at the decision time.

02 Establish required work

Open diagnostic concerns, then apply procedure eligibility and evidence-preservation rules.

03 Choose the next procedure

The candidate baseline uses a fixed rule priority. Helion ranks the same eligible procedures using seven initial fault probabilities.

The ML proposal changes the ranking step only.

Source: MOCK-ENG-002, decision boundary and evidence record. Design notebook, policy section.

Evidence update and closure

Each procedure changes the investigation state.

01 Run the procedure

The engineer approves the recommendation before the lab begins work.

02 Record the finding

Record each scoped result as confirmed present, confirmed absent or inconclusive.

03 Update and repeat

Refresh unresolved obligations and procedure eligibility, then recommend the next action.

04 Approve closure

The engineer closes the case only after the required diagnostic concerns have been resolved.

Source: MOCK-ENG-002, evidence record and stopping logic.

Business objective and diagnostic guardrail

Lower cost only counts at the same diagnostic standard.

01 Objective

Reduce the standard resources consumed per started investigation relative to the rule baseline.

02 Cost scope

Count engineer and technician time, equipment occupancy and procedure supplies.

03 Quality guardrail

Match the rule baseline on diagnostic completeness and false-negative performance.

04 Reporting meaning

Track turnaround separately. Resource dollars measure consumed capacity, not demonstrated avoidable cash.

Source: Client brief, Constraints and scope. SYN-OPS-001 cost-accounting assumptions.

From observations to one recommendation

01 Inputs

Use initial manufacturing, package-inspection and acceptance observations.

02 Model output

Produce seven independent fault probabilities so that mechanisms may coexist.

03 Decision policy

Combine probabilities with procedure coverage, inconclusive rates and standard resource costs.

04 Recommendation

Rank only the currently eligible procedures and present the next one for engineer review.

One model supports one next-test decision.

Source: Client brief, The Ask and Constraints and scope. Design notebook, policy definition.

Safety boundary and human control

01 Engineer authority

The engineer approves each procedure and the eventual closure decision.

02 No automatic absence

Low probability, an unexamined mechanism and an inconclusive result never confirm absence.

03 Coexisting faults

A first confirmed mechanism does not cancel other open diagnostic concerns.

04 Evidence updates

New findings change mechanism states and procedure eligibility. Initial model probabilities remain fixed.

The baseline and ML route retain the same diagnostic obligations.

Source: MOCK-ENG-002. Design notebook, policy and operating sections.